SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA0404

NAME: P.GAYATHRI REG.NO:192425277

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5 Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. **A coin is tossed 10 times and the outcomes are:**
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

- **Aim**
  To estimate the probability of getting Heads using the Maximum Likelihood Estimation (MLE) method.
  Formula

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

- Total tosses = 10
- Heads = 7
- Tails = 3
  Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

- **Code**

```
tosses = ['H','H','T','H','H','H','T','H','H','T']

heads = tosses.count('H')
total = len(tosses)

p = heads / total
```

```
print("Estimated Probability of Heads (MLE):", p)
```

- **Output**
  Estimated Probability of Heads (MLE): 0.7

- **Result**
  The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is **0.7**.

2. **Out of 50 students, 42 passed an exam.**
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

- **Aim**
  To estimate the probability that a student passes the exam using MLE.
  Formula

$$\hat{p} = \frac{\text{Students Passed}}{\text{Total Students}}$$

- **Calculation**

$$\hat{p} = \frac{42}{50} = 0.84$$

- **Code**

```
passed = 42
total = 50

p = passed / total

print("MLE of Passing Probability:", p)
```

- **Output**
  MLE of Passing Probability: 0.84

- **Result**
  The estimated probability that a student passes is **0.84**.

3. **A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.**
   Find the MLE for the probability that a bulb is defective.

- **Aim**
  To estimate the probability that a bulb is defective using MLE.
  Formula

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

- Total bulbs = 100
- Defective bulbs = 8
  Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

- **Code**

```
defective = 8
total = 100

p = defective / total

print("MLE of Defective Probability:", p)
```

- **Output**
  MLE of Defective Probability: 0.08

- **Result**
  The estimated probability that a bulb is defective is 0.08.

4. **A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.**

- **Aim**
  To estimate the probability of rolling a 6 using MLE.
  Formula

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

- Total rolls = 60
- Sixes = 18
  Calculation

$$\hat{p} = \frac{18}{60} = 0.3$$

Python Code

```
sixes = 18
rolls = 60

p = sixes / rolls

print("MLE of Rolling a 6:", p)
```

- **Output**
  MLE of Rolling a 6: 0.3

- **Result**
  The estimated probability of rolling a 6 is **0.3**.

5. **Given:**
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   y = w × x
   1. **Find the predicted output.**

2. **Calculate the error.**
3. **If the gradient is 4 and learning rate is 0.01, find the updated weight.**

- **Aim**

To calculate the predicted output, error, and updated weight using Gradient Descent.

Input (x) = 4
Weight (w) = 2
Actual Output = 10
Learning Rate = 0.01
Gradient = 4

Formula

Prediction:

$$y = w \times x$$

Error:

$$\text{Error} = \text{Actual Output} - \text{Prediction}$$

Weight Update:

$$w_{new} = w - \eta \times \text{Gradient}$$

where,

- η= Learning Rate

Calculation

Prediction:

$$2 \times 4 = 8$$

Error:

$$10 - 8 = 2$$

Updated Weight:

$$2 - (0.01 \times 4) = 1.96$$

Code

```
x = 4
w = 2
actual = 10
learning_rate = 0.01
gradient = 4

prediction = w * x
error = actual - prediction

updated_weight = w - learning_rate * gradient

print("Predicted Output:", prediction)
print("Error:", error)
```

```
print("Updated Weight:", updated_weight)
```

Output

```
Predicted Output: 8
Error: 2
Updated Weight: 1.96
```

Result

Predicted Output = 8
Error = 2
Updated Weight = 1.96

Code of Conduct:

I _P.GAYATHRI/192425277_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation